An AI-Based Intelligent Farming System for Improving Pomegranate Yield and Quality





ReeCha Farm



Akaran Farm



Mathu Farm

An AI-Based Intelligent Farming System for Improving Pomegranate Yield and Quality

StudentID	Proposal Comments	Action Taken by Team
IT22281296	No Comments	-
IT22549440	No Comments	-
IT22188304	Re-check confidence rate. Confidence rate showed as 99.9%.	Checked for data leakage, removed duplicates, retrained the model, and rechecked confidence scores. Convert to PDF
IT22245656	Need to be ready for the live demonstration.	Tested the application and practised the complete live demonstration.

An AI-Based Intelligent Farming System for Improving Pomegranate Yield and Quality

Problem

- Sri Lanka spends nearly 9000 million LKR annually on importing pomegranate [1] **EconomyNext**
- **Farmers still depend on traditional manual farming practices.**
- **Lack of knowledge about soil conditions, irrigation, and nutrient levels.**
- **Late disease detection causes crop damage and yield loss.**
- **No proper system to monitor fruit growth or predict harvest time.**
- **After harvesting, fruit grading is done manually, causing fruit waste.**

Impact of the Problem

Low farming productivity

Crop losses due to disease

Increased fruit imports

Financial loss for farmers





Users and Stakeholders

Pomegranate Farmers

Farm Owners

Agricultural Officers

Fruit Exporters

Agricultural Researchers

An AI-Based Intelligent Farming System for Improving Pomegranate Yield and Quality

Existing Solutions

- Many research systems focus on single agricultural problems only
- Leaf disease detection using image processing
- Soil monitoring systems measuring moisture, pH, and nutrients
- Fruit grading systems used after harvesting

Limitations

- Most disease detection systems analyze leaf diseases, not fruit diseases
- Soil monitoring systems only display parameters without recommendations
- No intelligent suggestions for fertilizer or irrigation
- Fruit growth stage monitoring is rarely considered
- Farmers still make manual decisions based on experience

Proposed Solutions

An integrated AI-based intelligent farming system that combines IoT sensors and image-based machine learning models to support farmers from soil monitoring to post-harvest fruit grading.

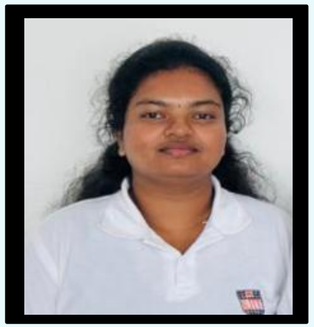
Our Team Member



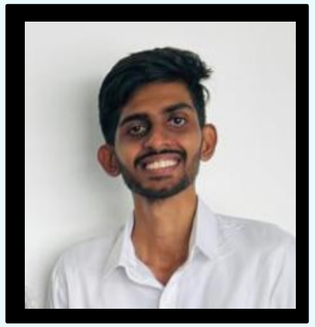
Team Leader
THISHOHARINI V
IT22281296



Member 02
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IT22549440



Member 03
BANUJA S
IT22188304



Member 04
KOBIRAM T
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IT22281296 | THISHOHARINI V

Fruit Growth Stage Detection and Harvest Time

An AI system that identifies the growth stage of a pomegranate from a single fruit image (Bud, Flower, Early Fruit, Mid-Growth or Mature Fruit) and predicts the time left to harvest.

Specialization: Information Technology

Individual Title: Fruit Growth Stage Detection and Harvest Time

Brief Description of the System

🔍 What the System Does

- Farmer uploads **one pomegranate image**.
- Identifies **five stages**: Bud, Flower, Early Fruit, Mid-Growth and Mature Fruit.
- **Rejects unrelated images**, such as apples, as Unknown.
- **Harvest prediction is rule-based**: the stage gives the estimated days to harvest, adjusted by weather.

Output classes

Bud | Flower | Early Fruit | Mid-Growth | Mature Fruit | Unknown

🧠 Two-Stage AI Pipeline

- **YOLO11s** detector (640 px) finds the fruit or flower in the photo.
- The detected region is cropped with 12% padding.
- **MobileNetV2** classifier labels the crop as one of 5 stages or Unknown.
- Low confidence returns **uncertain**; no detection asks the farmer to retake the photo.

YOLO11s · where is the fruit?

↓

MobileNetV2 · which stage?

↓

Rules + weather · days to harvest

📊 Training and Testing (80/20)

- **8,513 unique images:** 5,855 annotated pomegranate images and 2,658 unrelated images (other fruits, objects).
- **80% development / 20% final test**; 10% of development is held out for validation.
- Exact duplicates are removed and image groups kept together before splitting; augmentation comes **only afterwards**.



Train 6,168 (72%) · Validation 701 (8%) · Test 1,644 (19%) of 8,513 unique images.

Individual Title: Fruit Growth Stage Detection and Harvest Time

Results and Testimonials

Benchmark Tests: Accuracy and F1 Score

Final 80/20 pipeline (YOLO11s + MobileNetV2), held-out test set (1,423 images)

Overall accuracy	Bud	Flower	Early Fruit	Mid-Growth	Mature Fruit
97.6%	**95.0%** accuracy	**96.4%** accuracy	**92.6%** accuracy	**97.7%** accuracy	**99.5%** accuracy
5 stages + unrelated images	**F1 0.965**	**F1 0.971**	**F1 0.944**	**F1 0.969**	**F1 0.979**
1,423 test images	160 images	225 images	136 images	257 images	185 images

Accuracy per stage = share of its test images identified correctly (recall). Stage accuracy 96.6% · stage macro-F1 0.966 · unrelated images rejected 99.8% · genuine pomegranates wrongly rejected 0.4%.

Testing Findings

- **Apple test:** earlier models labelled green apples as Mid-Growth and red apples as Mature Fruit. After adding negative examples, 459 of 460 unrelated test images are rejected.
- **Early stages:** earlier models rejected genuine Buds and were weak on Flower and Early Fruit. Now Bud is 95.0% (1 of 160 rejected); Early Fruit, the weakest stage, is 92.6%.
- **Scope:** scores use 1,423 single-label test images; 221 images showing more than one stage are left out of this accuracy.

Individual Title: Fruit Growth Stage Detection and Harvest Time

Limitations and Future Improvements

⚠️ Limitations of the System

- **One fruit per photo:** the system cannot analyse a whole branch or count flowers and fruits.
- **Early stages are hardest:** Early Fruit scores lowest (92.6%), mostly confused with Mid-Growth; 0.4% of genuine pomegranate images are still rejected.
- **Public images only:** training, test and unrelated images are all public, so orchard accuracy is still to be confirmed.
- **Mixed-stage photos** (several stages in one image) are left out of the accuracy score; branch-level analysis will cover them.

🚀 Future Improvements: Branch-Level Analysis

1 Capture a branch

One photo shows every flower and fruit on the branch.

2 Analyse the image

The AI detects each flower and fruit and classifies its growth stage.

3 Branch report

- Number of flowers
- Number of small fruits
- Number of mature fruits
- Flowers to keep

- Today the system analyses **one fruit per photo**; the next version will analyse a **whole branch**.
- It will count **how many flowers** there are and **how many have turned into small fruits** on the branch.
- It will advise **how many flowers to keep** on each branch.
- Add more real-orchard Bud, Flower and Early Fruit photos, plus reviewed negatives (apples, bowls, other flowers).



IT22188304 | BANUJA S

Fruit Disease Detection & Treatment Recommendation

Identifies common diseases and problems affecting pomegranate plants and treatment recommendation.

Specialization: Information Technology

Individual Title: Fruit Disease Detection & Treatment Recommendation

💡 Feedback from users

1. Farmers found treatment guidance useful alongside disease detection.
2. One user highlighted its value for new farmers with little experience.
3. Most farmers preferred organic treatments because of cost and local market needs.
4. They encouraged further development, saying more guidance could attract more farmers.
5. They suggested adding at least five more treatments for each option.
6. One farmer offered to promote the app to other farmers through his social media.



“Treatment guidance is useful for new farmers.”

— Mr. Kunarasa Jeimathuran, Reecha Farm Supervisor —

Individual Title: Fruit Disease Detection & Treatment Recommendation

💡 Feedback from users

AGRICULTURAL EXPERT CONFIRMATION LETTER
AI-Based Smart Pomegranate Cultivation Support System
Final Year Research Project – Sri Lanka Institute of Information Technology (SLIIT)

Agricultural Instructor Confirmation and Declaration

I hereby certify that I have reviewed the agricultural information prepared by Banuja Sundareswaran of the Sri Lanka Institute of Information Technology (SLIIT), including the prevention tips, organic and chemical treatments, product formulations, dosages, application intervals, number of applications, pre-harvest intervals and follow-up guidance.

Items Reviewed and Confirmed
Please tick the items reviewed and confirmed as appropriate.

- ☑ The prevention tips are appropriate for local cultivation.
- ☑ The organic and chemical treatments are suitable for their recommended uses.
- ☑ The product formulations and quantities are correct.
- ☑ The application intervals and number of applications are appropriate.
- ☑ The pre-harvest intervals, precautions and follow-up guidance are suitable.

Additional Comments / Corrections / Conditions:
These recommendations are suitable for Sri Lanka but many vary with weather and field conditions.

I confirm that the items ticked above are agriculturally appropriate, practically applicable and suitable for inclusion in the student's Final Year Research Project application, subject to the corrections and conditions recorded above.

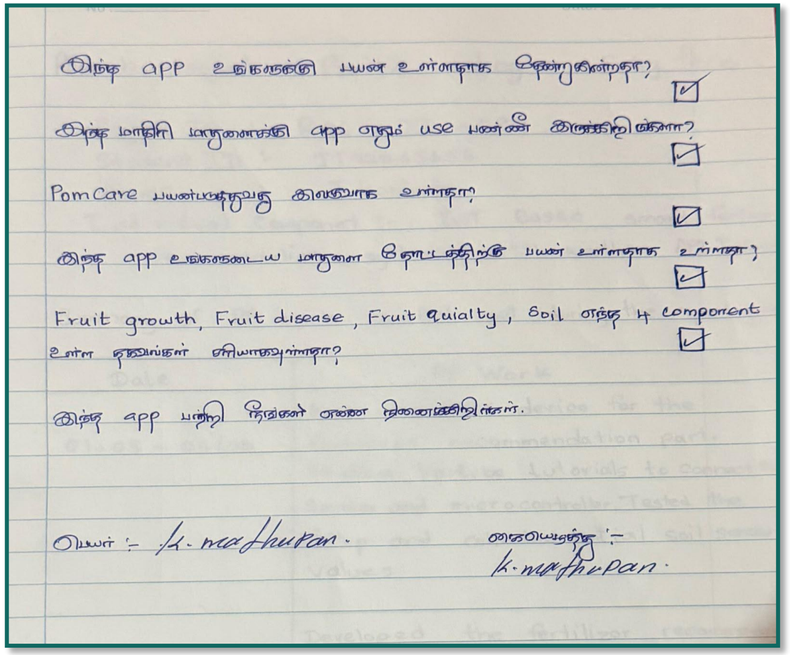
This confirmation is issued at the request of the student for academic purposes only.

Signature of Agricultural Instructor

Instructor's Name: M.Mahilan
Designation / Department: Agriculture Instructor
Date: Agriculture Extension Center
Official Stamp: Chunnakam, Jaffna

AGRICULTURAL EXPERT CONFIRMATION LETTER
AI-Based Smart Pomegranate Cultivation Support System
Final Year Research Project – Sri Lanka Institute of Information Technology (SLIIT)

 Copper oxychloride 50% WP (Moderate)	1.5 g	24 g	12 days	3	Remove infected fruits and improve airflow. Apply copper oxychloride 50% WP at 24 g per 16 L tank. Repeat every 12 days for three applications in total.
 Copper oxychloride 50% WP (Severe)	2 g	32 g	10 days	4	Remove heavily affected and rotting fruits promptly. Apply copper oxychloride 50% WP at 32 g per 16 L tank. Repeat every 10 days for four applications in total. Inspect nearby fruits for further infection.
BACTERIAL BLIGHT					
 Garlic bulb extract 30% (Mild)	300 ml extract + 700 ml water	4.8 L extract + 11.2 L water	14 days	2	Apply the prepared 30% garlic extract using 4.8 L extract and 11.2 L water for a 16 L tank. Repeat after 14 days for two applications in total. Monitor for new spots.
 Tulsi leaf extract 40%, then neem seed oil (Moderate)	400 ml extract + 600 ml water	6.4 L extract + 9.6 L water	10 days	2	Clear infected fruits and fallen debris. Spray the prepared 40% Tulsi extract using 6.4 L extract and 9.6 L water. Follow with a separate neem seed oil application at its label dose and timing. Repeat the sequence after 14 days.



"Treatment guidance is useful for new farmers."

— Mr. Kunarasa Jeimathuran, Reecha Farm Supervisor —

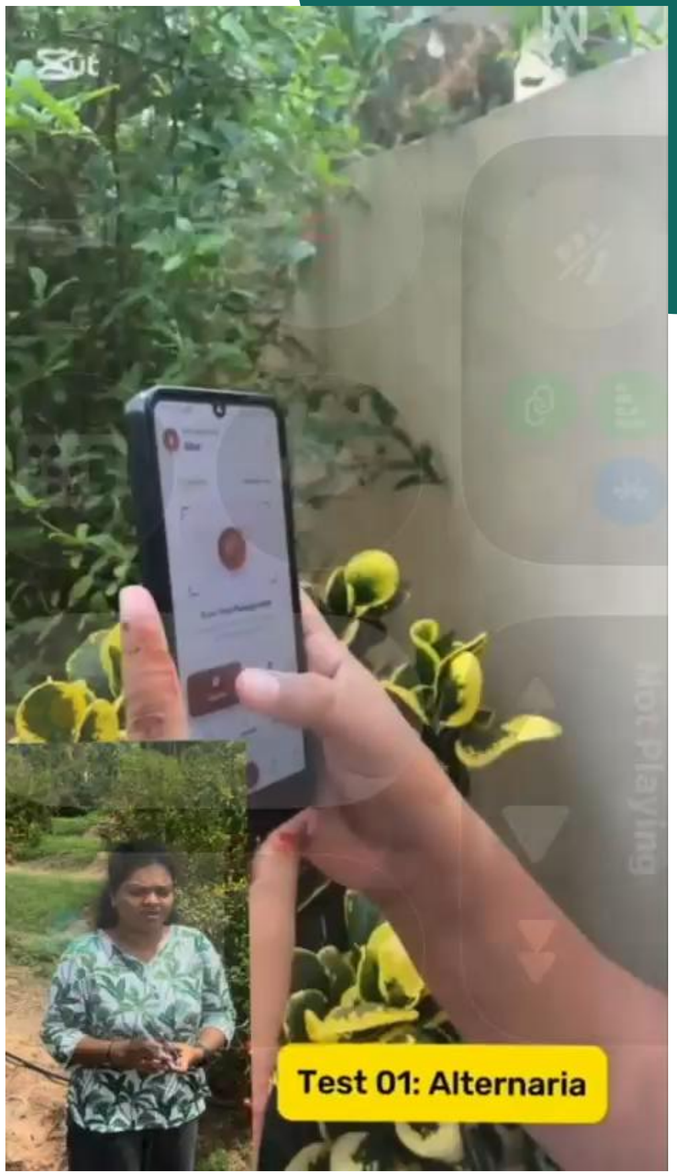
Individual Title: Fruit Disease Detection & Treatment Recommendation

💡 User Acceptance Test (UAT)

- ✅ Capture a fruit image
- ✅ Tap the Analyse button
- ✅ View the disease result
- ✅ View severity percentage and level
- ✅ View treatment guidance
- ✅ Choose a treatment option
- ✅ Set a treatment reminder

“Treatment guidance is useful for new farmers.”

— Mr. Kunarasa Jeimathuran, Reecha Farm Supervisor —



Individual Title: Fruit Disease Detection & Treatment Recommendation

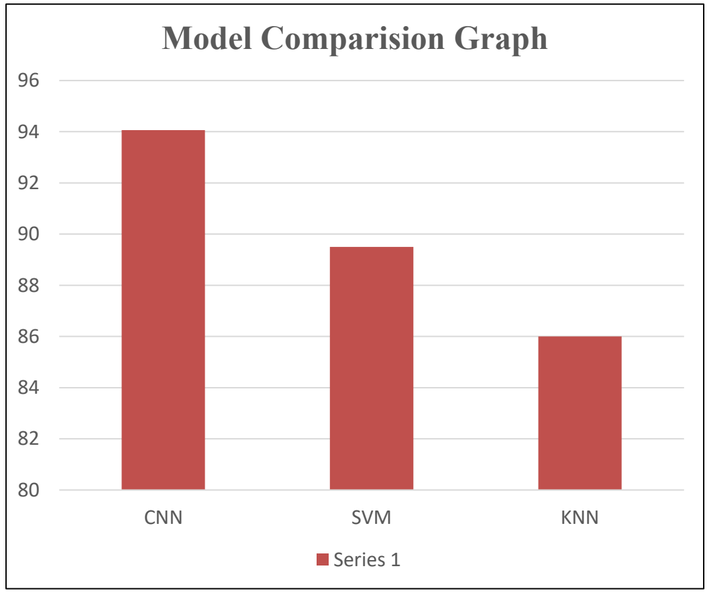
💡 RESULTS

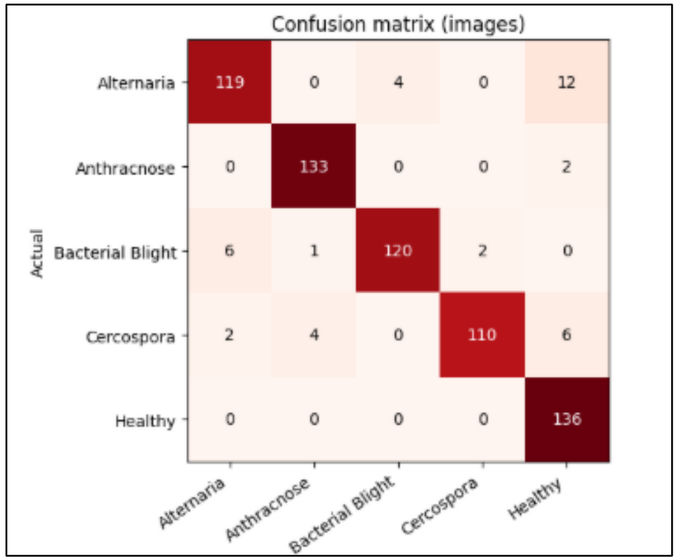
Model Comparision Table

DISEASE CLASSIFICATION MODEL COMPARISON

	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
0	CNN (MobileNetV2)	94.06	94.34	94.06	94.05
1	SVM	89.50	89.53	89.50	89.44
2	KNN	86.00	86.27	86.00	85.95

Model Comparision Graph





Confusion Matrix

Individual Title: Fruit Disease Detection & Treatment Recommendation

LIMITATIONS

- Poor photo quality and lighting can affect the results.
- Severity covers visible damage only, not internal or hidden damage.
- The system checks only one fruit per image, not multiple fruits at once.
- The system identifies only the four trained diseases.

FUTURE IMPROVEMENTS

- Collect more local farm images to improve disease detection.
- Support images from different angles to improve severity estimates.
- Add at least five more reviewed treatment suggestions for each treatment option.
- Use weather data from other components to support disease prevention guidance.



IT22549440 | SOWMIYA A

Fruit Quality Grading & Waste Utilization Analysis

Identifies the quality of pomegranate fruits using images to help farmers and recommend waste utilization

Specialization: Information Technology

Individual Title: Fruit Quality Grading & Waste Utilization Analysis

💡 Feedback from users

1. Users found quality grading helpful for checking fruit quality before buying or selling.
2. Farmers found the feature useful for sorting fruits according to their quality.
3. Users suggested extending the feature to supermarkets to help customers choose fruits.
4. One farm supervisor suggested adding internal quality checks to assess seed condition.
5. Farmers recommended focusing mainly on local market needs.



Individual Title: Fruit Quality Grading & Waste Utilization Analysis

💡 Feedback from users

Component 04 – Fruit Quality Grading and Waste Utilization Recommendation

Scope

Independently of disease status, the component grades the overall commercial quality of the fruit as High, Medium or Low, and recommends how the fruit should be utilized based on that grade – market sale for High and Medium quality, and waste or by-product utilization for Low quality.

Quality Categories

- High Quality – suitable for direct market or export sale
- Medium Quality – suitable for local market or processing use
- Low Quality – not suitable for sale; recommended for waste utilization such as composting, juice or by-product extraction, or discard as per standard practice

Method Used

Grading is based on a weighted scoring formula adapted from a standard published on pomegranate and fruit quality grading, combining multiple visual quality parameters such as size, colour uniformity, surface defect ratio and blemish coverage into a single weighted score. The resulting score is mapped to a High, Medium or Low grade using fixed cut-off thresholds, and a usage or waste-utilization recommendation is generated accordingly.

Items Reviewed and Confirmed

Item Reviewed and Confirmed	Confirmation
The reference paper's weighted parameter approach is a valid method for pomegranate quality grading.	✓
The chosen parameters and their relative weights are appropriate for grading commercial quality.	✓
The High / Medium / Low grade thresholds are reasonable dividing points.	✓
The usage recommendation (market sale) for High and Medium quality fruit is appropriate.	✓
The waste-utilization recommendation for Low quality fruit is appropriate and practical.	✓

Scanned with CamScanner

5. Expert Confirmation and Declaration

I hereby certify that I have reviewed the agricultural information presented in this document, including the selected parameters, recommended ranges, threshold values, stage-transition and harvest time estimates, disease and severity classifications, treatment recommendations, quality grading criteria and waste-utilization guidance.

I confirm that the data collected and used by the above-mentioned students of the Sri Lanka Institute of Information Technology (SLIIT) is agriculturally reasonable, practically applicable and suitable for use in their Final Year Research Project, and that all items listed in the confirmation tables above have been reviewed and confirmed as stated.

This confirmation is issued at the request of the students for academic purposes only.

Signature of the Agricultural Expert / Authorised Officer

Name: M.Mahilan

Designation / Position: Agriculture Instructor

Department / Office: Agriculture Extension Cen... Chunnakam, Jaffna

Contact Number: 0777045979

Email: mahilan22@gmail.com

Date:

Official Stamp of the Department / Office

Date: 15/10/2026

Farmer Feed back in Field Visits.

1. நாடளாவிய ரீதியிலான Supermarkets இல் தரம் பரிசோதித்தல்.
   Extend quality grading to supermarkets.
2. Customers இடம் பழங்களை வழங்குவதற்கு முன் அவற்றின் தரத்தை சரிபார்க்க உதவுதல்.
   Help customers check fruit quality before buying.
3. Seed தரத்தை சரிபார்க்க, பழத்தின் உட்புறத் தர மதிப்பீட்டை சேர்த்தல்.
   Add internal quality assesment to chebk seed condition
4. Users பழங்களை வாங்கவும் விற்கவும் இந்த App உதவியாக இருப்பதாக தெரிவித்தனர்.
   Users found quality grading helpful for buying and selling fruits ★★★★★

இந்த app உங்களுக்கு பயன் உள்ளதாக தென்படுகின்றதா? ✓

இந்த மாதிரி மாதுளைக்கு app ஏதும் use பண்ணி இருக்கிறீங்களா? ✓

PomCare பயன்படுத்துவது கிலையாக உள்ளதா? ✓

இந்த app உங்களுடைய மாதுளை தொடர்பிற்கு பயன் உள்ளதாக உள்ளதா? ✓

Fruit growth, Fruit disease, Fruit Quialty, Soil எந்த 4 component உள்ள தேவைப்பாடு சரியானவர்களா? ✓

இந்த app பற்றி நீங்கள் என்ன நினைக்கிறீங்கள்.

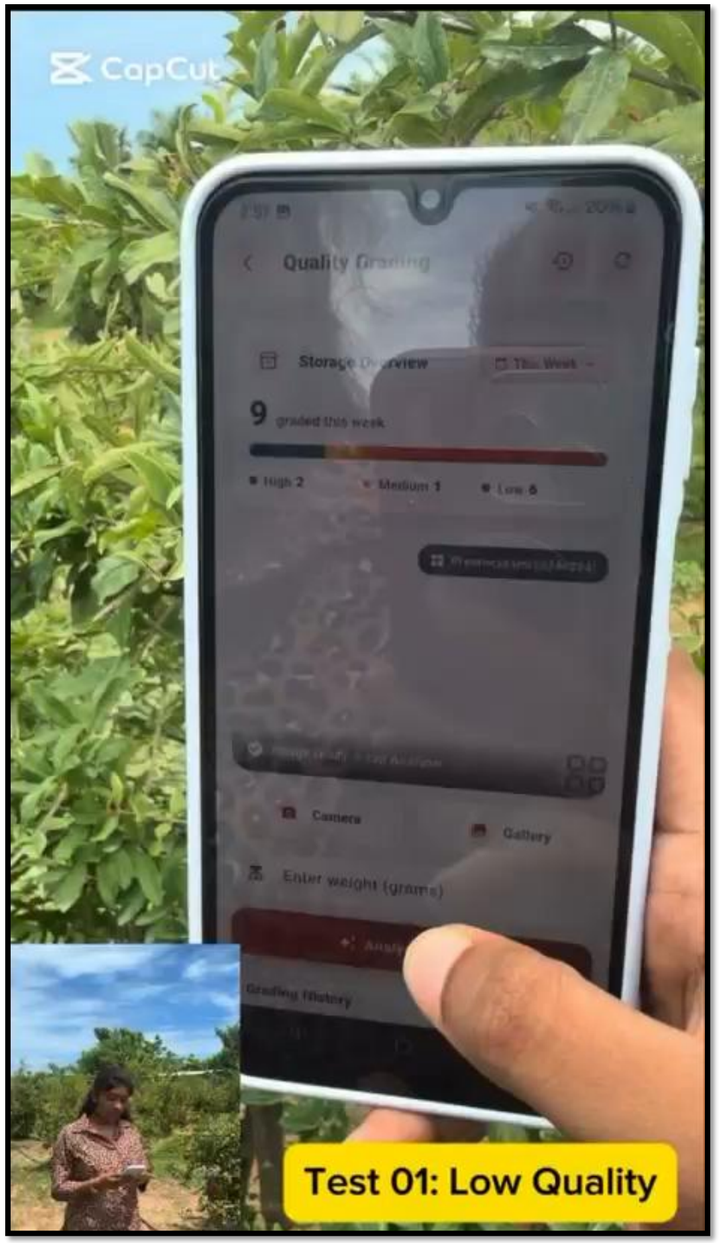
பெயர் :- K. mathuran.

கையெழுத்து :-
K. mathuran.

Individual Title: Fruit Quality Grading & Waste Utilization Analysis

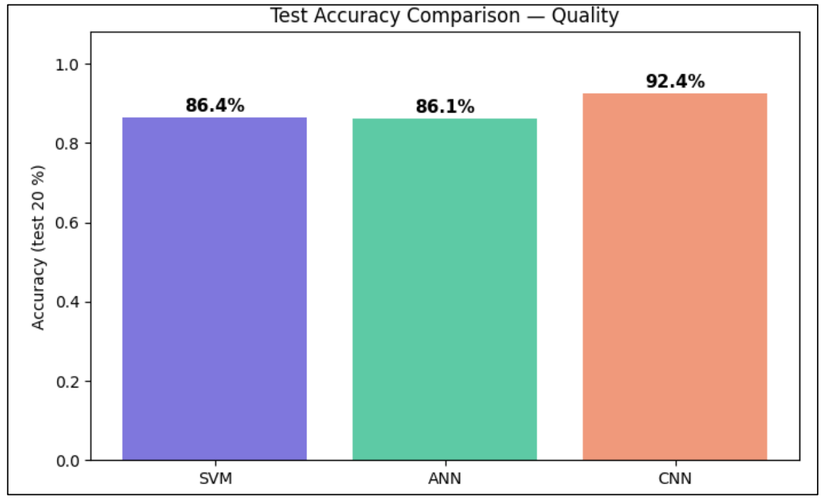
💡 User Acceptance Test (UAT)

- ✅ Capture a fruit image
- ✅ Enter the weight
- ✅ Tap the Analyse button
- ✅ View the quality result
- ✅ View defect type and level
- ✅ View usage recommendation



Individual Title: Fruit Quality Grading & Waste Utilization Analysis

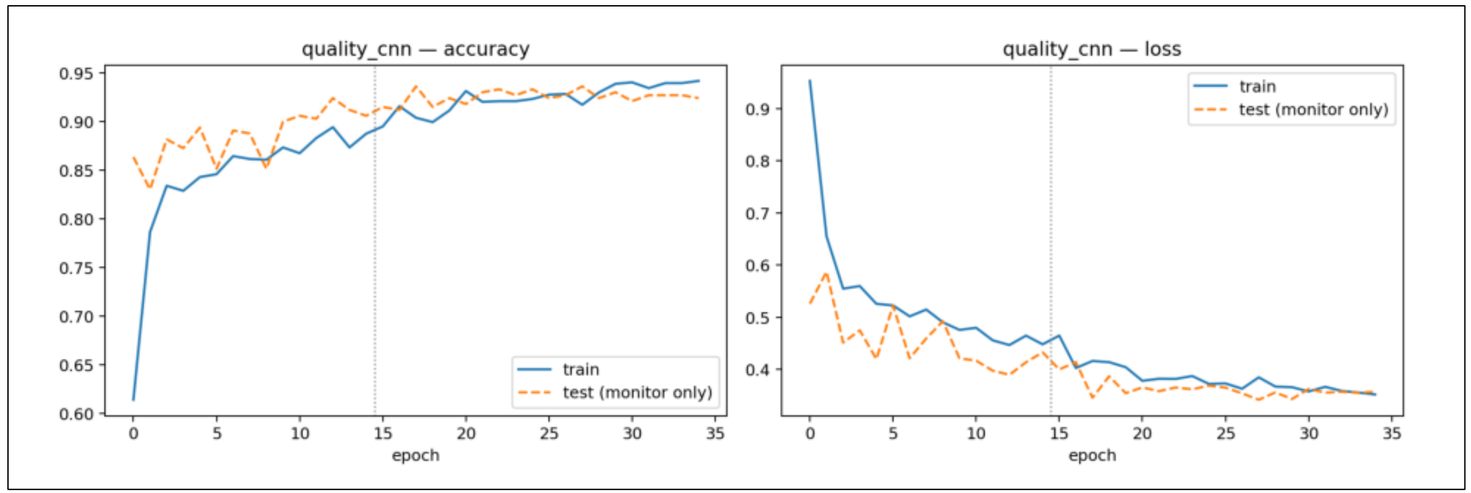
Comparison Graph

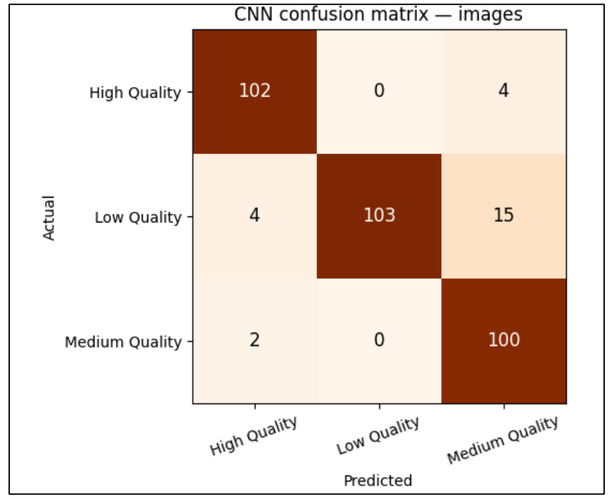


Test Accuracy Comparison — Quality

model	Accuracy	Precision	Recall	F1-score	Test images
SVM	86.4%	86.3%	86.6%	86.4%	330
ANN	86.1%	86.1%	86.3%	86.1%	330
CNN	92.4%	92.8%	92.9%	92.5%	330

Confusion Matrix





No Overfitting

Individual Title: Fruit Quality Grading & Waste Utilization Analysis

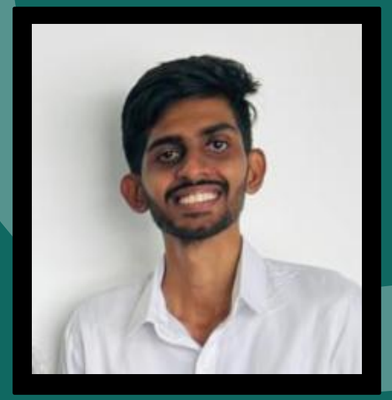
LIMITATIONS

- Checks only visible external quality, not internal seed condition.
- Poor lighting and image quality may affect grading results.
- Defects on the hidden side of the fruit may be missed.
- Results may vary for fruit varieties not included in training.

FUTURE IMPROVEMENTS

- Add internal quality assessment using suitable technology.
- Support grading multiple fruits at once.
- Train with more local fruit images under different lighting conditions.
- Extend the feature to supermarkets and improve guidance for local market needs.

IT22549440 | SOWMIYA A | R26-IT-087



IT22245656 | KOBIRAM T

Intelligent Fertilizer Recommendation & Irrigation Decision System

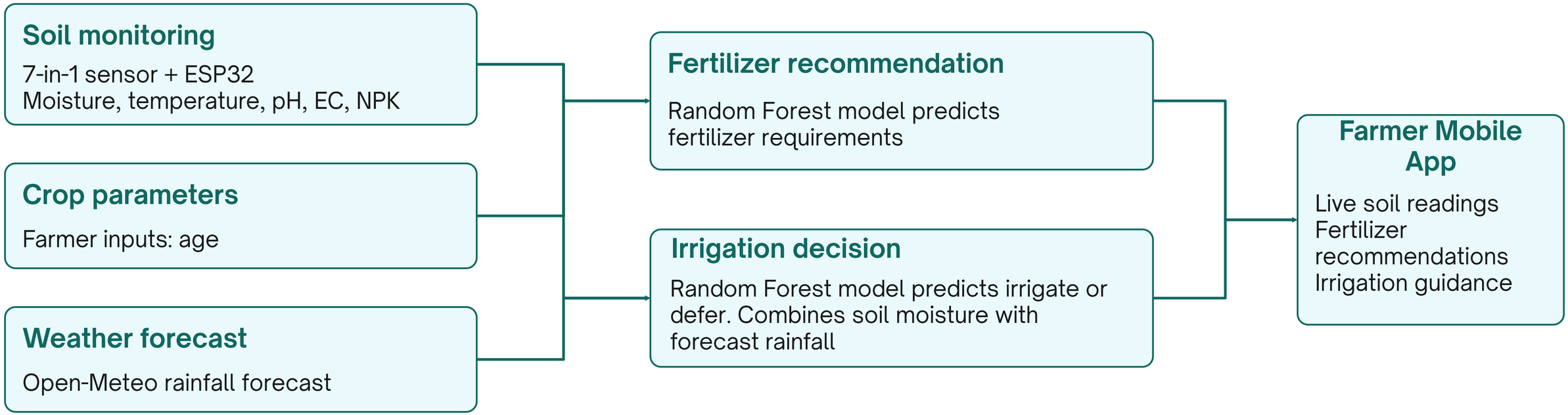
Combines soil sensor data, tree information, and weather forecasts to help farmers use fertilizer and water efficiently.

Specialization: Information Technology

Individual Title: Intelligent Fertilizer Recommendation & Irrigation Decision System

An IoT and machine learning system for pomegranate cultivation that monitors soil conditions, recommends fertilizer requirements and uses rainfall forecasts to guide irrigation decisions.

System Overview



IT22245656 | KOBIRAM T | R26-IT-087

Individual Title: Intelligent Fertilizer Recommendation & Irrigation Decision System

💡 Feedback from users

1. Real-time soil readings help us monitor moisture and nutrient conditions in the field.
2. Fertilizer recommendations help us plan nutrient application for pomegranate plants.
3. Rainfall forecasts help us decide when to irrigate and avoid unnecessary watering.
4. Local-language instructions would make the recommendations easier to understand.
5. Low-moisture alerts and offline access would improve usability in the field.

IT22245656 | KOBIRAM T | R26-IT-087

Individual Title: Intelligent Fertilizer Recommendation & Irrigation Decision System

LIMITATIONS

-

FUTURE IMPROVEMENTS

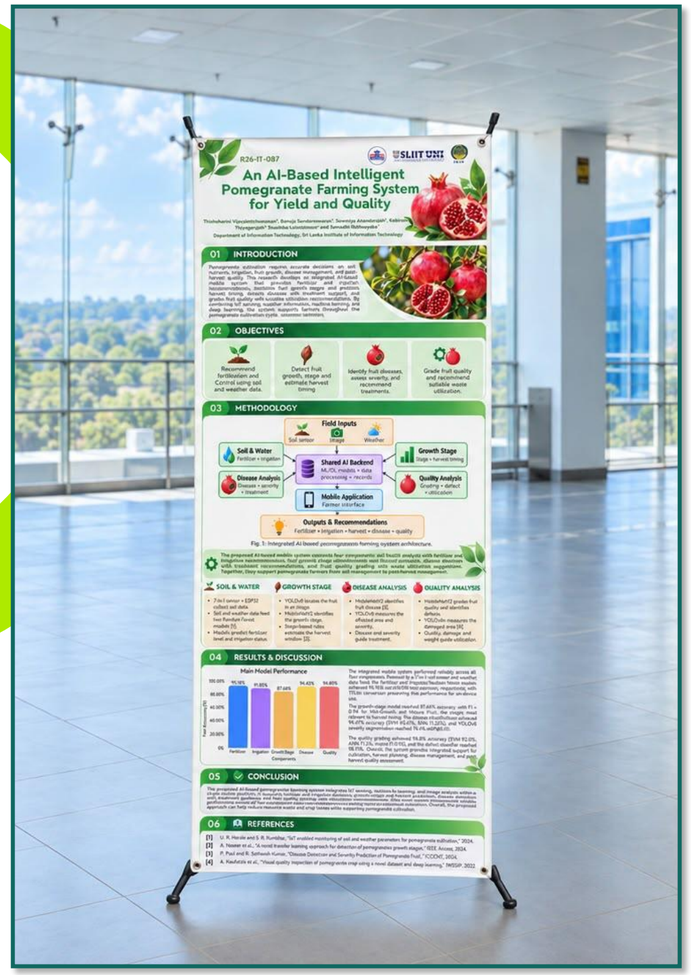
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IT22245656 | KOBIRAM T | R26-IT-087

An AI-Based Intelligent Farming System for Improving Pomegranate Yield and Quality



An AI-Based Intelligent Farming System for Improving Pomegranate Yield and Quality





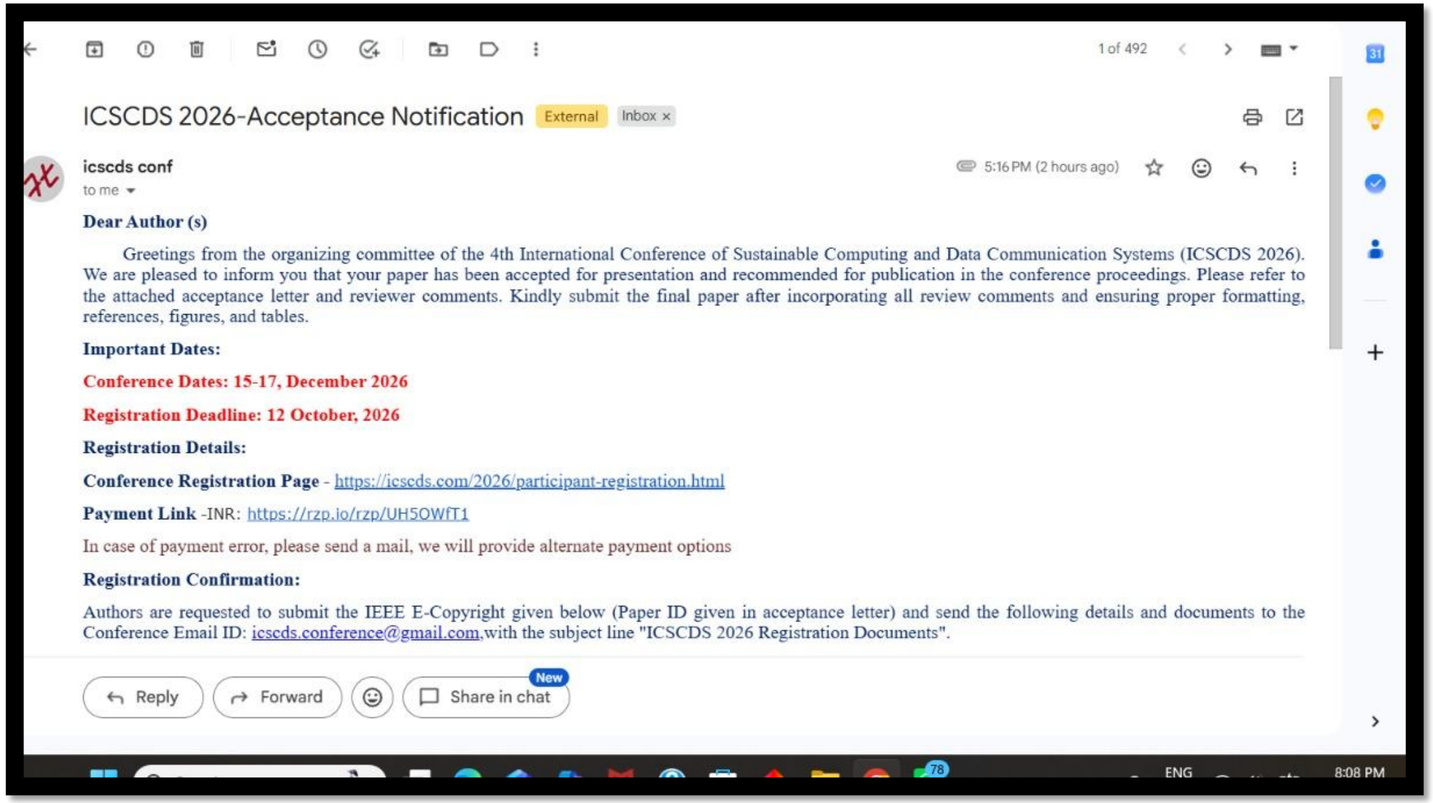
🏆 Runner-Up – ICDMIS 2026 🥈





An AI-Based Intelligent Farming System for Improving Pomegranate Yield and Quality

Acceptance Letter



ICSCDS 2026-Acceptance Notification External Inbox ×

1 of 492

icscds conf
to me
5:16 PM (2 hours ago)

Dear Author (s)

Greetings from the organizing committee of the 4th International Conference of Sustainable Computing and Data Communication Systems (ICSCDS 2026). We are pleased to inform you that your paper has been accepted for presentation and recommended for publication in the conference proceedings. Please refer to the attached acceptance letter and reviewer comments. Kindly submit the final paper after incorporating all review comments and ensuring proper formatting, references, figures, and tables.

Important Dates:

Conference Dates: 15-17, December 2026

Registration Deadline: 12 October, 2026

Registration Details:

Conference Registration Page - https://icscds.com/2026/participant-registration.html

Payment Link -INR: https://rzp.io/rzp/UH5OWfT1

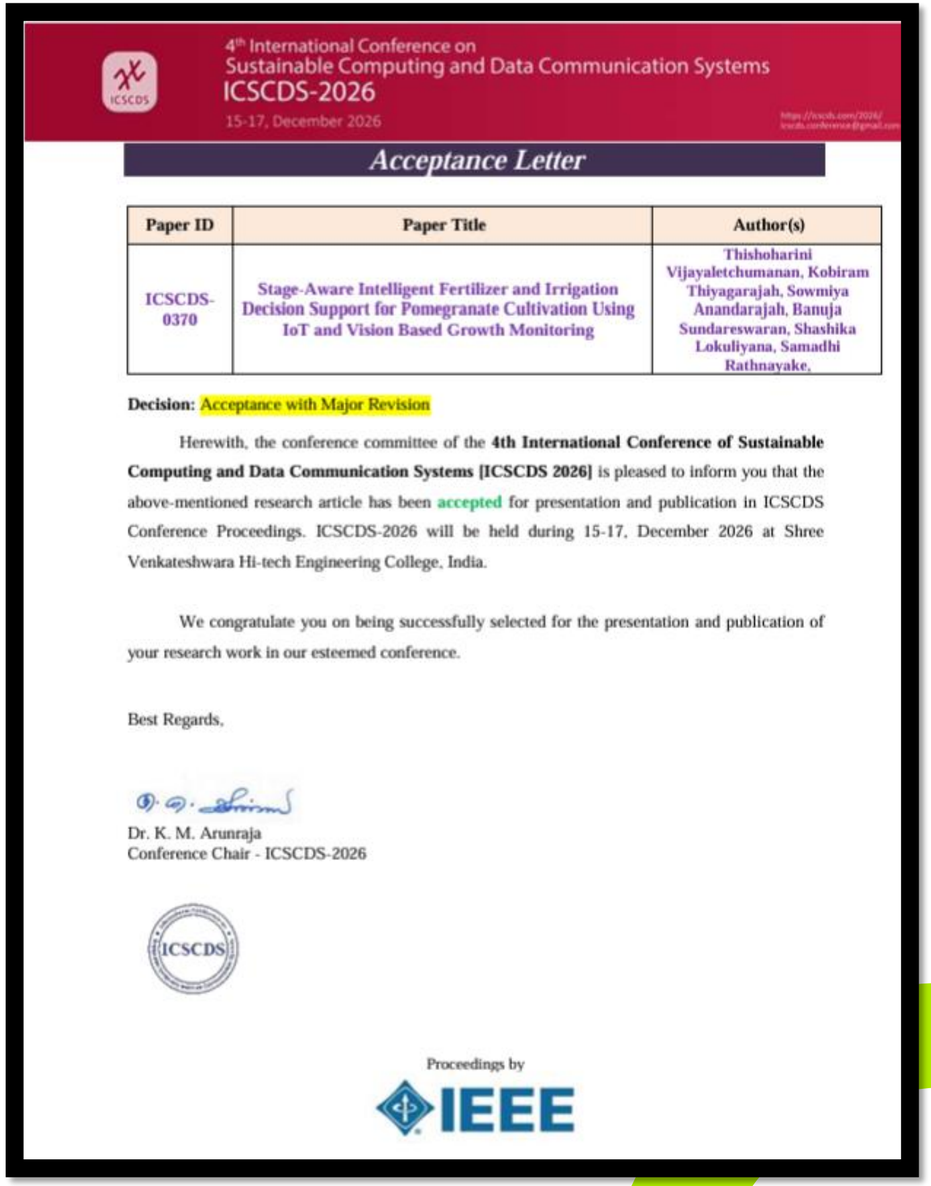
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Registration Confirmation:

Authors are requested to submit the IEEE E-Copyright given below (Paper ID given in acceptance letter) and send the following details and documents to the Conference Email ID: icscds.conference@gmail.com,with the subject line "ICSCDS 2026 Registration Documents".

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4th International Conference on
Sustainable Computing and Data Communication Systems
ICSCDS-2026
15-17, December 2026

Acceptance Letter

Paper ID	Paper Title	Author(s)
ICSCDS-0370	Stage-Aware Intelligent Fertilizer and Irrigation Decision Support for Pomegranate Cultivation Using IoT and Vision Based Growth Monitoring	Thishoharini Vijayaletchumanan, Kobiram Thiyagarajah, Sowmiya Anandarajah, Banuja Sundareswaran, Shashika Lokuliyana, Samadhi Rathnayake,

Decision: Acceptance with Major Revision

Herewith, the conference committee of the **4th International Conference of Sustainable Computing and Data Communication Systems [ICSCDS 2026]** is pleased to inform you that the above-mentioned research article has been accepted for presentation and publication in ICSCDS Conference Proceedings. ICSCDS-2026 will be held during 15-17, December 2026 at Shree Venkateshwara Hi-tech Engineering College, India.

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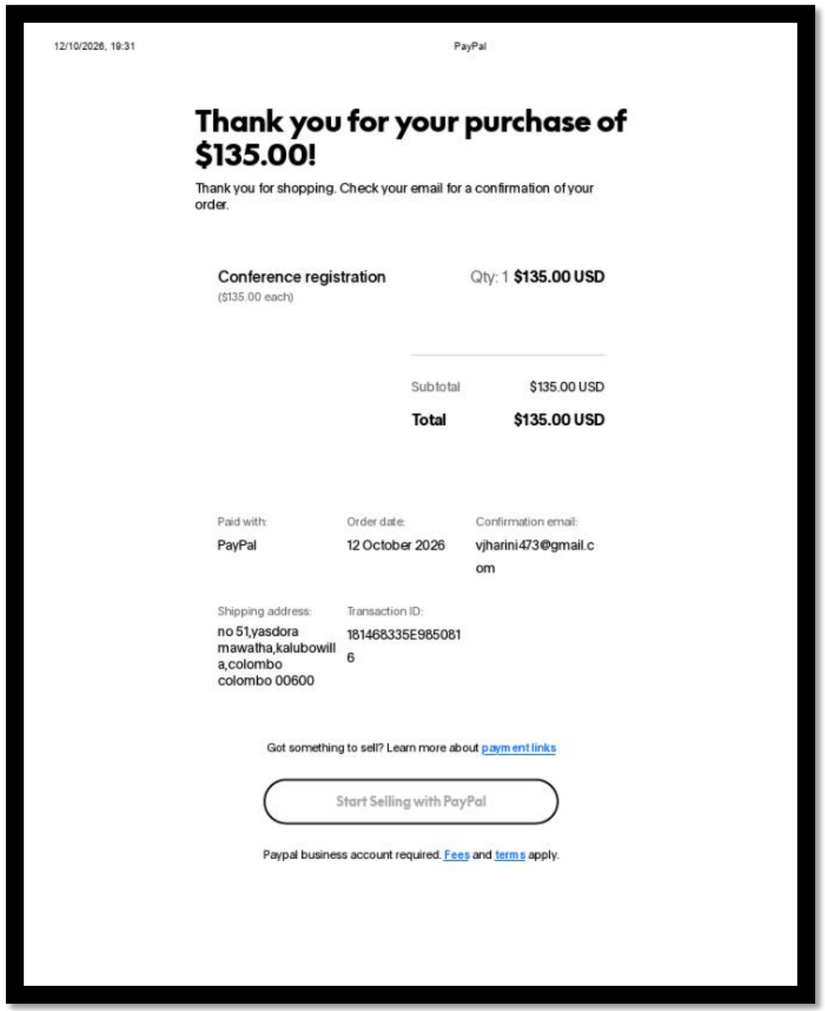
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Stage-Aware Intelligent Fertilizer and Irrigation Decision Support for Pomegranate Cultivation Using IoT and Vision Based Growth Monitoring

Thisheharini Vijayaletchumanan,Kobiram Thiyagarajan,Bunja Sundareswaran,Sewmiya Anandarajah,Shashika Lokuliyana,Samadhi Rathnayake

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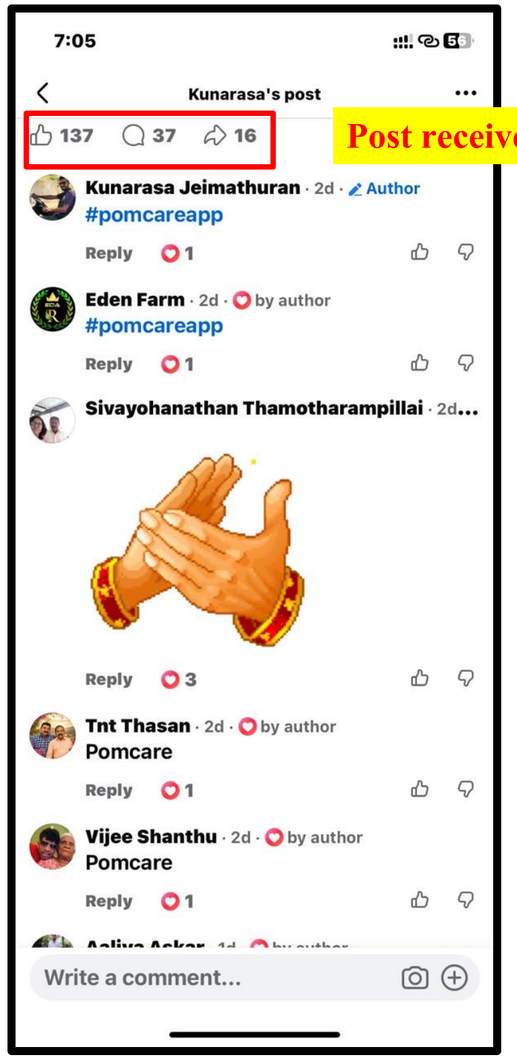
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